

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0612

Commonwealth Chesapeake 345 kV
November 01, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0612. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Commonwealth Chesapeake 345 kV substation in the Virginia area. The request was received on April 24, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 260 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$154,557,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of SOUTHTOWN – COMMONWEALTH CHESAPEAKE SUBSTATION 345.0 kV L, the VEDDER – OTAY LAKE 138.0 kV loads to 365.1% of its 248 MVA rating. Project DFAX: 47.0%.

COMMONWEALTH CHESAPEAKE 138-1 kV Transformer Bank: 359.4% loading (464 MVA) following the outage of the COMMONWEALTH CHESAPEAKE – SOUTHTOWN 345.0 kV Line; DFAX = 100.0%.

HUNTINGTON FALLS – OTAY LAKE 138.0 kV: 358.6% loading (254 MVA) following the outage of the COMMONWEALTH CHESAPEAKE – SOUTHTOWN 345.0 kV line; DFAX = 47.0%.

For the loss of SOUTHTOWN – COMMONWEALTH CHESAPEAKE 345.0 kV, the COMMONWEALTH CHESAPEAKE 345-1 kV TX loads to 295.8% of its 564 MVA rating. Project DFAX: 100.0%.

The CANDELARIA ARENAS 1718 – OCEAN CITY SUBSTATION 138.0 kV L is overloaded to 289.7% of its 268 MVA emergency rating for the loss of the VIENNA OPERATIONS – SOUTHTOWN 345.0 kV L. The project contributes 45.3% DFAX (678.8 MW) to this overload.

For the loss of SOUTHTOWN – VIENNA OPERATIONS 345.0 kV, the OCEAN CITY – REIDLAND 138.0 kV L loads to 269.9% of its 283 MVA rating. Project DFAX: 45.3%.

For the loss of SOUTHTOWN – VIENNA OPERATIONS 345.0 kV, the ARROYO TAP – REIDLAND 138.0 kV Line loads to 269.9% of its 283 MVA rating. Project DFAX: 45.3%.

For the loss of SOUTHTOWN SUBSTATION – COMMONWEALTH CHESAPEAKE 345.0 kV, the DANE – COMMONWEALTH CHESAPEAKE 138.0 kV loads to 260.6% of its 222 MVA rating. Project DFAX: 29.0%.

For the loss of SOUTHTOWN – COMMONWEALTH CHESAPEAKE 345.0 kV, the COLCHESTER – QUINCE ORCHARD SUBSTATION 138.0 kV loads to 260.3% of its 186 MVA rating. Project DFAX: 29.0%.

The VEDDER – LANGLOIS 138.0 kV is overloaded to 258.1% of its 185 MVA emergency rating for the loss of the COMMONWEALTH CHESAPEAKE – SOUTHTOWN 345.0 kV. The project contributes 27.4% DFAX (410.5 MW) to this overload.

HUNTINGTON FALLS – COMMONWEALTH CHESAPEAKE 138.0 kV: 250.6% loading (260 MVA) following the outage of the SOUTHTOWN SUBSTATION – COMMONWEALTH CHESAPEAKE 345.0 kV; DFAX = 34.6%.

MESQUITE – ARROYO TAP 138.0 kV line: 247.6% loading (302 MVA) following the outage of the SOUTHTOWN – VIENNA OPERATIONS 345.0 kV Line; DFAX = 45.3%.

The COLCHESTER – VALLEYVIEW 138.0 kV is overloaded to 242.7% of its 210 MVA emergency rating for the loss of the SOUTHTOWN – COMMONWEALTH CHESAPEAKE 345.0 kV. The project contributes 29.0% DFAX (434.5 MW) to this overload.

The N LAKE – HITCHLAND 138.0 kV line is overloaded to 226.7% of its 227 MVA emergency rating for the loss of the VIENNA OPERATIONS – EASTON 2 345.0 kV Line. The project contributes 27.2% DFAX (408.1 MW) to this overload.

SHINGLE CRK – SOUTHTOWN 138.0 kV: 220.9% loading (370 MVA) following the outage of the SOUTHTOWN – VIENNA OPERATIONS 345.0 kV; DFAX = 54.7%.

For the loss of SOUTHTOWN – COMMONWEALTH CHESAPEAKE 345.0 kV, the LANGLOIS SUBSTATION – SOUTHTOWN SUBSTATION 138.0 kV loads to 219.6% of its 206 MVA rating. Project DFAX: 27.4%.

For the loss of SOUTHTOWN – VIENNA OPERATIONS SUBSTATION 345.0 kV, the MESQUITE – RIPPLE 138.0 kV loads to 212.9% of its 337 MVA rating. Project DFAX: 45.3%.

SOUTHTOWN 345-138 kV Autotransformer: 207.4% loading (452 MVA) following the outage of the SOUTHTOWN – VIENNA OPERATIONS 345.0 kV; DFAX = 55.2%.

HITCHLAND – SHERWOOD 138.0 kV: 206.3% loading (236 MVA) following the outage of the EASTON 2 – VIENNA OPERATIONS 345.0 kV line; DFAX = 27.2%.

For the loss of SOUTHTOWN – VIENNA OPERATIONS 345.0 kV, the SANTEE LAH – SUNSET 138.0 kV L loads to 197.1% of its 233 MVA rating. Project DFAX: 22.2%.

W -STATION – SHINGLE CREEK 138.0 kV L: 195.6% loading (403 MVA) following the outage of the SOUTHTOWN – VIENNA OPERATIONS 345.0 kV; DFAX = 54.7%.

For the loss of VIENNA OPERATIONS – EASTON 2 345.0 kV, the SANDBURG – SHERWOOD 138.0 kV loads to 191.5% of its 251 MVA rating. Project DFAX: 27.2%.

KIWI – N LAKE 138.0 kV: 190.3% loading (241 MVA) following the outage of the EASTON 2 – VIENNA OPERATIONS 345.0 kV; DFAX = 24.7%.

For the loss of EASTON 2 – VIENNA OPERATIONS SUBSTATION 345.0 kV, the SANDBURG – JARRATT SUBSTATION 138.0 kV loads to 188.0% of its 251 MVA rating. Project DFAX: 27.2%.

For the loss of SOUTHTOWN – VIENNA OPERATIONS SUBSTATION 345.0 kV, the SANTEE LAH – GALVANIZING 138.0 kV line loads to 186.9% of its 242 MVA rating. Project DFAX: 22.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VEDDER – OTAY LAKE 138.0 kV	Rebuild line with higher-capacity conductor	\$5,871,000	74% (\$4,344,000)
COMMONWEALTH CHESAPEAKE 138-1 kV Autotransformer	Replace transformer with higher-rated unit	\$3,460,000	90% (\$3,108,000)
HUNTINGTON FALLS – OTAY LAKE 138.0 kV	Rebuild line with higher-capacity conductor	\$8,800,000	74% (\$6,511,000)
COMMONWEALTH CHESAPEAKE 345 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$23,730,000	90% (\$21,321,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CANDELARIA ARENAS 1718 – OCEAN CITY SUBSTATION 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,281,000	73% (\$1,669,000)
OCEAN CITY – REIDLAND 138.0 kV L	Rebuild line with higher-capacity conductor	\$8,614,000	73% (\$6,303,000)
ARRYO TAP – REIDLAND 138.0 kV line	Rebuild line with higher-capacity conductor	\$850,000	73% (\$622,000)
DANE – COMMONWEALTH CHESAPEAKE 138.0 kV L	Rebuild line with higher-capacity conductor	\$18,897,000	75% (\$14,127,000)
COLCHESTER SUBSTATION – QUINCE ORCHARD 138.0 kV	Rebuild line with higher-capacity conductor	\$4,109,000	75% (\$3,072,000)
VEDDER – LANGLOIS SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$17,349,000	74% (\$12,812,000)
HUNTINGTON FALLS – COMMONWEALTH CHESAPEAKE 138.0 kV	Rebuild line with higher-capacity conductor	\$2,939,000	76% (\$2,236,000)
MESQUITE – ARROYO TAP 138.0 kV	Rebuild line with higher-capacity conductor	\$2,350,000	73% (\$1,720,000)
COLCHESTER – VALLEYVIEW 138.0 kV	Rebuild line with higher-capacity conductor	\$12,568,000	75% (\$9,396,000)
HITCHLAND – NLAKE 138.0 kV	Rebuild line with higher-capacity conductor	\$12,924,000	70% (\$9,072,000)
SOUTHTOWN – SHINGLE CRK 138.0 kV	Rebuild line with higher-capacity conductor	\$7,728,000	74% (\$5,713,000)
LANGLOIS – SOUTHTOWN 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,604,000	74% (\$4,877,000)
MESQUITE – RIPPLE 138.0 kV	Rebuild line with higher-capacity conductor	\$3,903,000	73% (\$2,856,000)
SOUTHTOWN 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$16,920,000	79% (\$13,342,000)
HITCHLAND – SHERWOOD 138.0 kV	Rebuild line with higher-capacity conductor	\$5,623,000	70% (\$3,947,000)
SANTEETLAH – SUNSET 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,579,000	70% (\$3,919,000)
W -STATION – SHINGLE CREEK 138.0 kV L	Rebuild line with higher-capacity conductor	\$10,474,000	74% (\$7,743,000)
SHERWOOD – SANDBURG 138.0 kV	Rebuild line with higher-capacity conductor	\$3,584,000	70% (\$2,516,000)
KIWI – N LAKE 138.0 kV	Rebuild line with higher-capacity conductor	\$4,116,000	71% (\$2,919,000)
SANDBURG – JARRATT SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$8,175,000	70% (\$5,739,000)
SANTEETLAH – GALVANIZING 138.0 kV	Rebuild line with higher-capacity conductor	\$6,651,000	70% (\$4,673,000)
	Total allocated to project		\$154,557,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.